

Information Overload under Rational Learning^{*}

Jonghyun Park[†]

Sudipta Sarangi[‡]

Hector Tzavellas[§]

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Abstract

A larger signal set need not improve the accuracy of rational inference when the cost of implementing a data-responsive rule increases with the dimension of the information environment. We study an agent who observes a vector of signals and chooses an inference rule to minimize mean squared error plus a processing cost. The cost depends on the responsiveness of the induced estimate and may rise with ambient dimension even when the same lower-dimensional estimate remains statistically reproducible. If every sequence of rules with nonvanishing responsiveness becomes prohibitively costly as dimension grows, the optimal estimate converges to the prior, and its accuracy returns to the prior benchmark. We also show that the optimal rule is affine when the posterior mean is affine. In a Gaussian–quadratic benchmark, the optimal rule is the Bayesian posterior mean shrunk by a dimension-dependent cognitive deflator. Finally, the same shrinkage can make the behavioral and payoff consequences of correlation neglect vanish in large information environments without making the misspecified covariance model correct.

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[†]Department of Economics, Virginia Tech. E-mail: jongh2419@vt.edu

[‡]Department of Economics, Virginia Tech. E-mail: ssarangi@vt.edu

[§]Department of Economics, Virginia Tech. E-mail: hectortz@vt.edu

1 Introduction

Data-rich environments are now standard in financial analysis, management, regulation, consumer choice, and digital communication. A larger set of indicators can improve statistical inference by providing more measurements of the underlying state. At the same time, producing a judgment that responds appropriately to a growing collection of inputs can require greater cognitive, organizational, or computational resources. The economic value of additional information therefore depends not only on what the signals contain, but also on how the decision maker transforms them into an estimate.

A broad empirical and experimental literature documents that decision quality need not improve monotonically with information load. Larger information sets can reduce choice quality, alter search and cue-utilization strategies, and weaken responsiveness to individual pieces of information (Eppler and Mengis, 2004; Roetzel, 2019).¹ Even information that is irrelevant under a simple decision rule may enter information integration, increase decision time, and create demand for less complex environments (Chadd et al., 2021; Dummel et al., 2016). These findings are commonly described as *information overload*: accuracy initially benefits from additional information but eventually deteriorates as the information environment becomes sufficiently large.

This pattern presents a basic challenge for economic theory. Suppose that information environments are nested, so that every signal available in a smaller environment remains available in a larger one. If a previously feasible inference rule can be implemented at the same cost after additional signals arrive, the decision maker can apply that rule to the original coordinates and ignore the new ones. The feasible set expands, and optimized statistical accuracy cannot decline. This monotonicity logic underlies standard Bayesian decision theory (Blackwell, 1953) and is preserved in many models of information acquisition, rational inattention, and selective response when unused inputs do not change the cost of an otherwise unchanged strategy (Gabaix, 2014; Grossman and Stiglitz, 1980; Matějka and McKay, 2015; Sims, 2003; Vives, 1993).

This paper studies optimal inference when that cost invariance fails. At information dimension d , an agent has access to the full signal vector $S^{(d)}$ and chooses a measurable rule ℓ that maps the observed signals into an estimate $\ell(S^{(d)})$ of an unknown state. She minimizes mean squared error plus a processing cost. The endogenous object is therefore not the sig-

¹Consumer-choice experiments document lower choice quality or fewer correct selections under larger information sets (Jacoby et al., 1974; Malhotra, 1982). In accounting environments, greater information load can reduce the consistency and quality of judgments (Chewning Jr and Harrell, 1990). In social media, exposure to a larger volume of content is associated with weaker responsiveness to each individual exposure (Rodriguez et al., 2014).

nal structure or the subset of signals observed. It is the rule used to transform an already available vector of signals into an estimate.

The processing cost depends on the induced estimate and on the ambient dimension in which the estimate is produced. We measure the rule’s responsiveness by $\text{Var}(\ell(S^{(d)}))$. A constant estimate has zero responsiveness, while an estimate that varies with the realized signals has positive responsiveness. Responsiveness is not itself a measure of statistical quality: an estimate can respond to either information or noise. Instead, it measures the extent to which the decision maker’s judgment changes across realizations of the information environment.

Ambient dimensionality may affect the resources required to maintain such responsiveness. An analyst working with a larger reporting system may need to reconcile more inputs, maintain a more complex inference procedure, or devote greater attention to ensuring that the estimate reacts appropriately to the available data. A rule that produces a particular estimate from the original coordinates can therefore become more costly when that rule is implemented within a larger information system. We treat this dimension dependence as a reduced-form property of the processing technology and characterize its consequences for optimal inference.

The decision maker remains fully optimizing. She knows the correct joint distribution of the state and signals, observes all available signals, and chooses the learning rule that minimizes her total expected loss. Information overload is not generated by a mechanical limit on the number of signals she can observe or by an exogenously imposed mistake in Bayesian updating. It arises because statistical responsiveness is valuable but costly, and the cost of maintaining a given degree of responsiveness can increase with the size of the surrounding information environment.

The central mechanism follows from the availability of a costless no-learning option. Under squared-error loss, the prior mean is the best constant estimate. The agent can therefore eliminate the processing cost of responsiveness by ignoring the realized signals and using the prior mean. At moderate dimensions, the accuracy gained from reacting to informative signals can justify a responsive rule. As dimensionality grows, however, the cost of maintaining that response can outweigh its statistical benefit. The agent then optimally attenuates the effect of the signals and places greater weight on the prior. More information can consequently produce a less responsive and less accurate estimate even though every lower-dimensional estimate remains statistically reproducible.

Our first contribution is to characterize this mechanism without imposing a parametric signal structure or a particular functional form for the learning rule. We assume that constant rules are costless and that any sequence of learning rules retaining nonvanishing respon-

siveness eventually becomes prohibitively costly as dimensionality increases. Under these conditions, the optimal induced estimate converges in mean square to the prior mean, its responsiveness converges to zero, and its mean squared error converges to the prior variance. We define information overload as the existence of a finite information dimension whose accuracy strictly exceeds the accuracy attained at every richer dimension. Within the class of processing technologies considered here, overload occurs if and only if learning produces a strict accuracy gain at some finite dimension. Useful information is therefore necessary for overload because if no finite information environment improves on the prior, there is no learning gain that can subsequently be lost.

Our second contribution concerns the structure of optimal inference. The general problem is infinite-dimensional because the agent chooses among all measurable functions of the signal vector. We show that this problem reduces to a finite-dimensional coefficient choice when two conditions hold. First, the Bayesian posterior mean is affine in the primitive signals. Second, processing costs are weakly increasing in the variance of the induced estimate. Under these conditions, the optimal rule is affine.

We then study a Gaussian-quadratic benchmark. The state is Gaussian, each signal is an independent noisy measurement of the state, and the processing cost equals the product of the information dimension, a cost parameter, and the variance of the induced estimate. The optimal rule is the Bayesian posterior mean shrunk toward the prior mean by a simple cognitive deflator.

A larger information environment then produces two opposing forces. Averaging more independent signals improves statistical precision and makes the Bayesian posterior more responsive to the sample average. At the same time, the cognitive deflator decreases with dimension and suppresses the agent’s optimal response. The first force dominates at low dimensions, while the second eventually dominates. Mean squared error is therefore single-troughed in information dimension: accuracy initially improves and subsequently deteriorates toward the prior benchmark. This produces the inverted-U accuracy pattern associated with information overload.

We distinguish this accuracy criterion from the agent’s total loss. Information overload concerns the mean squared error of the optimal rule at a given exogenous dimension. When the agent may instead choose the dimension itself, she minimizes mean squared error plus processing cost. We derive the exact preferred information dimension and show how it depends on signal noise, prior uncertainty, and the processing-cost coefficient. Greater signal noise increases the value of averaging and raises the preferred number of signals, while a higher processing cost or greater prior variance reduces it. We also extend the benchmark to

heterogeneous signal quality. Processing costs shrink all coefficients through a common cognitive deflator while preserving the relative Bayesian weights assigned to signals of different precision.

Finally, we examine how dimension-dependent processing costs affect are related to correlation neglect (Enke and Zimmermann, 2019). We take a misspecified covariance model as given and compare the rule it generates with the optimal rule under the true covariance structure. When positively correlated signals are treated as independent, the correlation-neglect rule responds too strongly to redundant information. With positive processing costs, however, both the correlation-aware and correlation-neglect rules are increasingly shrunk toward the prior as dimensionality grows. Their induced estimates converge to one another, and the true total-loss penalty from correlation neglect converges to zero.

The paper’s main implication is that rationality alone does not guarantee that a larger information environment produces a more informative decision. The relevant benchmark also requires that a previously available degree of responsiveness remain equally inexpensive to implement. Once the cost of responsiveness depends on ambient dimensionality, a rational agent may optimally react less to an expanding collection of informative signals. Information overload is then an endogenous feature of optimal inference rather than a direct consequence of irrationality or exogenous observation constraints.

Related Literature

The paper first contributes to the empirical and experimental literature on information overload. Studies in consumer choice, accounting, organizational decision making, and information systems document that larger information sets can change search behavior, reduce cue utilization, lengthen decision time, and lower decision quality; see Eppler and Mengis (2004) and Roetzel (2019) for reviews. Early consumer-choice experiments show that decision quality may decline as the number of available attributes or alternatives increases (Jacoby et al., 1974; Malhotra, 1982). Related work finds that information load affects processing strategies and the number of cues incorporated into judgments (Chewning Jr and Harrell, 1990; Payne, 1976). Particularly relevant for our mechanism, irrelevant information may not be behaviorally neutral. Dummel et al. (2016) show that formally irrelevant information can enter information integration, while Chadd et al. (2021) find that irrelevant information increases decision time, contributes to suboptimal choices, and creates demand for simpler environments. We represent these processing consequences through a dimension-dependent cost of responsiveness and study the inference rule chosen by an optimizing decision maker.

The analysis is also related to Bayesian decision theory, costly information acquisition, ratio-

nal inattention, and sparse decision making. In the Blackwell framework, a more informative experiment weakly lowers Bayes risk because every decision rule available under a less informative experiment can be reproduced under the more informative one (Blackwell, 1953). Costly information-acquisition models allow agents to choose how much information to obtain (Grossman and Stiglitz, 1980; Vives, 1993), while rational-inattention models attach costs to selected information structures or posterior distributions (Caplin and Dean, 2015; Matějka and McKay, 2015; Sims, 2003). Sparsity-based models generate selective response by penalizing complex representations or sensitivity to multiple variables (Gabaix, 2014).

Our choice object is different. The signal vector is given and fully observed, and the agent chooses the mapping from those signals into an estimate. The processing cost depends on the responsiveness of the induced estimate and on the dimension of the environment in which it is implemented. Consequently, the same estimate can remain statistically feasible while becoming more costly to produce in a larger environment. This distinction allows optimized statistical accuracy to decline even though the underlying signal sets are nested.

Several economic models generate overload-like outcomes through search, attention allocation, or costly extraction of relevant content. Branco et al. (2016) study information provision when consumers conduct costly sequential search, while Lu (2022) shows that more detailed financial reports can make relevant information harder for rationally inattentive investors to extract. Blankespoor et al. (2020) review a broader literature in which monitoring, acquiring, and analyzing public disclosures are costly.² These models place the friction in the acquisition or extraction of information. We instead study the implementation of an inference rule conditional on a given signal vector. The formulation isolates the extent to which ambient dimensionality changes the optimal responsiveness of the final estimate.

The paper is distinct from the literature on choice overload and limited consideration. Choice-overload studies ask how a larger menu of actions affects participation, satisfaction, or choice quality (Iyengar and Lepper, 2000). Models of limited consideration allow the decision maker to evaluate only a subset of the available alternatives (Lleras et al., 2017; Masatlioglu et al., 2012). In our setting, the expanding objects are signals rather than actions. The action or estimation problem is fixed, and dimensionality changes the set of informational inputs available for forming an estimate. The relevant margin is therefore the responsiveness of inference, not the composition of a consideration set over alternatives.

Finally, the correlation-neglect exercise connects the paper to research on correlated information and learning under misspecified models. Enke and Zimmermann (2019) provide exper-

²Related work studies how disclosure presentation, timing, and competition from simultaneous announcements affect investor attention and market responses (DellaVigna and Pollet, 2009; Hirshleifer et al., 2009; Hirshleifer and Teoh, 2003).

imental evidence that individuals underadjust for correlation across signals, and Dasaratha and He (2021) study learning when information sources are correlated or redundant. A broader theoretical literature examines inference and behavior under misspecified statistical models (Dasaratha and He, 2020; Esponda and Pouzo, 2016; Frick et al., 2020; Spiegler, 2016). We take the covariance misspecification as given and identify how a separate processing friction changes its behavioral and payoff implications. The correlation-aware and correlation-neglect rules become similar because both are shrunk toward the prior, not because the subjective covariance model converges to the truth.

The rest of the paper is organized as follows. Section 2 presents the statistical environment, learning problem, and processing technology. Section 3 establishes learning collapse and characterizes information overload. Section 4 derives conditions under which the optimal learning rule is affine. Section 5 studies the Gaussian–quadratic benchmark, the cognitive deflator, and the preferred information dimension. Section 6 examines the consequences of correlation neglect. Section 7 concludes.

2 Model

This section develops a model of inference under dimension-dependent processing costs. At each information dimension, the signal vector is given and fully observed. The agent’s choice is not which signals to acquire, but how to transform their realized values into an estimate of the state. We separate the statistical content of the signal vector from the cost of implementing a responsive inference rule. The dimension d indexes the size of the information environment, with $d + 1$ signals available at dimension d . It should not be interpreted as a universal measure of cognitive or computational complexity. Whether a larger environment is more burdensome is determined by the processing technology introduced below.

2.1 Information Environment

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space supporting a real-valued state θ and a sequence of real-valued signals $(S_k)_{k \geq 0}$. The state has a finite, nondegenerate second moment:

$$\mu_\theta := \mathbb{E}[\theta], \quad 0 < \sigma_\theta^2 := \text{Var}(\theta) < \infty.$$

At information dimension $d \in \mathbb{N}_0$, the agent observes

$$S^{(d)} := (S_0, S_1, \dots, S_d)^\top \in \mathbb{R}^{d+1}.$$

Uppercase letters denote random variables, while $s^{(d)} := (s_0, s_1, \dots, s_d)^\top$ denotes a realization of $S^{(d)}$. Unless otherwise stated, all expectations, variances, and covariances are evaluated under the joint law $\mathbb{P}$ of the state and signals.³

We initially impose no parametric restriction on the joint distribution of $(\theta, S^{(d)})$. The signals may be biased, correlated, heterogeneous in quality, redundant, partly informative, or completely uninformative. The formulation includes repeated noisy measurements $S_k = \theta + \varepsilon_k$, reports with heterogeneous noise variances, signals affected by common sources, and environments containing both informative and irrelevant coordinates.⁴

The information environments are nested. Every coordinate available at dimension d remains available at every dimension $d' > d$, and an increase in d leaves the joint distribution of the previously available state and signals unchanged. A larger d therefore expands the statistical information available to the agent without removing any previously feasible use of the original coordinates.

2.2 Learning Rules and Processing Costs

At dimension d , a learning rule is a Borel-measurable function

$$\ell : \mathbb{R}^{d+1} \rightarrow \mathbb{R}$$

that maps the realized signal vector into an estimate $\ell(S^{(d)})$ of θ . Let

$$\mathcal{L}^{(d)} := \{ \ell : \mathbb{R}^{d+1} \rightarrow \mathbb{R} : \ell \text{ is Borel measurable and } \mathbb{E}[\ell(S^{(d)})^2] < \infty \}.$$

We identify two learning rules whenever they induce the same estimate almost surely.⁵ The processing technology respects this identification. If $\ell, \tilde{\ell} \in \mathcal{L}^{(d)}$ satisfy

$$\ell(S^{(d)}) = \tilde{\ell}(S^{(d)}) \quad \text{almost surely,}$$

then

$$C(\ell; d) = C(\tilde{\ell}; d).$$

³Equivalently, one may begin with a projectively consistent family of finite-dimensional joint laws for $(\theta, S^{(d)})$. The common-probability-space representation makes the nesting of the information environments explicit.

⁴One may interpret S_0 as the agent's own signal and $S_1, \dots, S_d$ as reports received from additional sources. None of the general results depends on this interpretation.

⁵Square integrability ensures that both the mean squared error and the variance of the induced estimate are well-defined.

The economically relevant object is therefore the induced estimate rather than a particular functional representation of the rule.

We refer to

$$\text{Var}(\ell(S^{(d)}))$$

as the *responsiveness* of the learning rule. Responsiveness measures how much the agent's estimate varies across realizations of the information environment. A constant rule has zero responsiveness, while a rule that reacts nontrivially to the observed signals generally has positive responsiveness. Responsiveness is not itself a measure of accuracy or informativeness: an estimate may vary because it responds to useful information, to statistical noise, or to both.

Nestedness ensures that every lower-dimensional estimate remains statistically reproducible in a richer environment. Fix $d' > d$ and $\ell \in \mathcal{L}^{(d)}$. Define the lifted rule

$$\ell_{d'}^+(s_0, \dots, s_{d'}) := \ell(s_0, \dots, s_d).$$

The lifted rule ignores the additional coordinates and satisfies

$$\ell_{d'}^+(S^{(d')}) = \ell(S^{(d)}) \quad \text{almost surely.}$$

It therefore generates exactly the same estimate, mean squared error, and responsiveness as the original rule. At the level of statistical feasibility, a richer information environment never eliminates a previously available inference strategy.

The implementation cost of that strategy, however, may depend on the environment in which it is used. At dimension d , the processing cost of a learning rule is

$$C(\ell; d) \in [0, +\infty].$$

The value $+\infty$ permits cognitive, organizational, or technological constraints to be represented as prohibitive costs. Although the cost depends only on the induced estimate at a fixed dimension, it may also depend directly on d . Consequently, the lifted rule need not preserve implementation cost:

$$C(\ell_{d'}^+; d') \neq C(\ell; d)$$

in general. The same statistical output may therefore require different resources when it is produced within information systems of different sizes.

At each dimension d , the agent chooses a learning rule to minimize mean squared error plus

processing cost:

$$\inf_{\ell \in \mathcal{L}^{(d)}} \left\{ \mathbb{E} \left[\left(\ell(S^{(d)}) - \theta \right)^2 \right] + C(\ell; d) \right\}. \quad (P_d)$$

The first term measures statistical inaccuracy. The second measures the resources required to implement the corresponding degree of responsiveness in an environment of dimension d .

We impose two substantive restrictions on the processing technology. The first provides a costless benchmark that does not respond to the realized signals. The second describes how the cost of sustaining responsiveness behaves as the surrounding information environment expands.

Assumption 1. *For every $d \geq 0$ and every $c \in \mathbb{R}$, if $\ell(s^{(d)}) \equiv c$, then $C(\ell; d) = 0$.*

Assumption 1 gives the agent a costless no-learning option. Under squared-error loss, the best such option is the prior-mean rule $\ell(s^{(d)}) \equiv \mu_\theta$. The agent can therefore avoid all responsiveness costs by reporting the prior mean independently of the realized signals.

Assumption 2. *For every sequence of learning rules $\{\ell_d\}_{d \geq 0}$ with $\ell_d \in \mathcal{L}^{(d)}$, if*

$$\liminf_{d \rightarrow \infty} \text{Var}(\ell_d(S^{(d)})) > 0,$$

then

$$\lim_{d \rightarrow \infty} C(\ell_d; d) = +\infty.$$

Assumption 2 states that nonvanishing responsiveness cannot be sustained at uniformly bounded cost as the information environment becomes arbitrarily large. It does not require the cost of a fixed rule to increase after every addition of a signal, nor does it require processing costs to be monotone in d at all finite dimensions. It imposes only an asymptotic restriction: every sequence of induced estimates whose variance remains bounded away from zero must eventually become prohibitively costly.

The assumption permits the lifted version of a fixed lower-dimensional rule to become more costly in a larger environment. Depending on the application, implementing a responsive estimate may require maintaining a larger reporting system, reconciling additional inputs, monitoring interactions among variables, verifying that excluded cues remain irrelevant, or allocating attention across a broader set of potential considerations. Even when the final estimate responds directly to only a subset of the coordinates, preserving that response within a larger information system may require additional cognitive or organizational resources. Evidence that larger cue sets alter information-processing strategies and that irrelevant information can increase decision time motivates this reduced-form dimension dependence (Chadd et al., 2021; Dummel et al., 2016; Payne, 1976). Assumption 2 does not derive these

costs from a primitive model of attention, search, or organizational design. It characterizes the consequences of a processing technology under which maintaining a nontrivial response becomes increasingly demanding as the environment expands.

2.3 Well-Posedness

To establish existence and uniqueness of the optimal estimate, fix $d \geq 0$ and let

$$\mathcal{H}_d := L^2(\Omega, \sigma(S^{(d)}), \mathbb{P})$$

denote the Hilbert space of square-integrable estimates measurable with respect to the observed signal vector. Its inner product and norm are

$$\langle X, Y \rangle_d := \mathbb{E}[XY], \quad \|X\|_2 := \sqrt{\mathbb{E}[X^2]}.$$

Every learning rule $\ell \in \mathcal{L}^{(d)}$ induces an estimate $X = \ell(S^{(d)}) \in \mathcal{H}_d$. Conversely, because $S^{(d)}$ takes values in the standard Borel space $\mathbb{R}^{d+1}$, every $X \in \mathcal{H}_d$ can be represented as $X = \ell(S^{(d)})$ for some Borel-measurable function ℓ . Since rules that induce the same estimate almost surely are identified, problem (P_d) is equivalently a choice over $\mathcal{H}_d$.

For $X = \ell(S^{(d)})$, define

$$C_d(X) := C(\ell; d).$$

The objective becomes

$$J_d(X) := \mathbb{E}[(X - \theta)^2] + C_d(X), \quad X \in \mathcal{H}_d.$$

Assumption 3. *For every $d \geq 0$, the functional $C_d : \mathcal{H}_d \rightarrow [0, +\infty]$ is:*

- (i) proper: $C_d(X) < +\infty$ for at least one $X \in \mathcal{H}_d$;
- (ii) convex: for every $X, Y \in \mathcal{H}_d$ and every $\alpha \in [0, 1]$,

$$C_d(\alpha X + (1 - \alpha)Y) \leq \alpha C_d(X) + (1 - \alpha)C_d(Y);$$

- (iii) lower semicontinuous: if $X_n \rightarrow X$ in L^2 , then

$$C_d(X) \leq \liminf_{n \rightarrow \infty} C_d(X_n).$$

Properness rules out a processing technology under which every estimate has infinite cost.

Under Assumption 1, properness is automatic because every constant estimate has zero cost. Convexity means that combining two induced estimates does not cost more than the corresponding convex combination of their costs. Lower semicontinuity rules out upward jumps in cost at limits of nearby estimates.

The following result shows the existence of a unique estimate of the decision problem.

Proposition 1. *Under Assumption 3, problem (P_d) admits a unique optimal induced estimate*

$$X_d^* = \ell_d^*(S^{(d)}) \in \mathcal{H}_d$$

for every $d \geq 0$, where uniqueness is understood up to almost-sure equality.

We may consequently define the statistical accuracy of optimal inference without selecting among multiple optimal estimates:

$$\text{MSE}(d) := \mathbb{E} \left[\left(\ell_d^*(S^{(d)}) - \theta \right)^2 \right].$$

A lower value of $\text{MSE}(d)$ represents greater statistical accuracy. This object should be distinguished from the minimized value of problem (P_d) , which also includes the processing cost of implementing the optimal rule.

Assumption 1 provides a dimension-independent benchmark. The prior-mean rule satisfies

$$\mathbb{E}[(\mu_\theta - \theta)^2] + C(\mu_\theta; d) = \sigma_\theta^2.$$

Optimality therefore implies

$$\text{MSE}(d) + C(\ell_d^*; d) \leq \sigma_\theta^2.$$

Because processing costs are nonnegative,

$$\text{MSE}(d) \leq \sigma_\theta^2 \quad \text{for every } d \geq 0. \tag{1}$$

Thus, the agent can always attain the prior benchmark by declining to respond to the observed signals. The substantive question is whether she chooses a more accurate responsive estimate and whether any such accuracy gain survives as dimensionality increases.

3 Information Overload

Because the implementation cost of a learning rule may depend on ambient dimension, the statistical nesting of the signal vectors does not determine the path of optimal accuracy. A

richer environment preserves every lower-dimensional estimate as a mathematical mapping, but it may change the cost of maintaining that estimate or any other responsive rule. This section establishes two implications of the processing technology.

The first is *prior reversion*: under Assumption 2, the optimal estimate converges to the prior mean as dimensionality grows. The second is *information overload*: a finite information dimension produces an estimate that is strictly more accurate than the optimal estimate at every richer dimension. Overload therefore requires both a finite-dimensional gain from learning and the eventual erosion of that gain.

By Proposition 1, the optimal induced estimate is uniquely defined at each dimension under Assumption 3. Recall that

$$\text{MSE}(d) = \mathbb{E} \left[\left(\ell_d^*(S^{(d)}) - \theta \right)^2 \right].$$

Information overload is defined in terms of this statistical accuracy, rather than the full objective in problem (P_d) .

Definition 1. *Information overload occurs if there exists a finite $d_1 \geq 0$ such that*

$$\text{MSE}(d) > \text{MSE}(d_1)$$

for every $d > d_1$.

Definition 1 requires more than diminishing marginal returns to information. It requires a finite-dimensional environment whose optimal estimate is strictly more accurate than the optimal estimate in every richer environment. The definition does not require $\text{MSE}(d)$ to increase after each successive addition of a signal. Accuracy may be locally flat or nonmonotone before and after d_1 . In the Gaussian-quadratic benchmark later, we obtain the sharper single-troughed path commonly associated with information overload.

We first establish the asymptotic behavior of the optimal estimate.

Proposition 2. *Suppose Assumptions 1, 2, and 3 hold. Then*

$$\lim_{d \rightarrow \infty} \mathbb{E} \left[\left(\ell_d^*(S^{(d)}) - \mu_\theta \right)^2 \right] = 0. \tag{2}$$

Consequently,

$$\text{Var}(\ell_d^*(S^{(d)})) \rightarrow 0, \quad \mathbb{E}[\ell_d^*(S^{(d)})] \rightarrow \mu_\theta,$$

and

$$\lim_{d \rightarrow \infty} \text{MSE}(d) = \sigma_\theta^2. \tag{3}$$

The proposition identifies the limiting response to an expanding information environment. It does not require the optimal rule to become exactly constant at any finite dimension. Instead, the induced estimate converges in mean square to the prior mean. Its variation across signal realizations vanishes, its expectation approaches the prior mean, and its accuracy returns to the level attained by ignoring the signals altogether.

The economic force behind the result is the interaction of the two cost assumptions. Because the prior-mean rule is always available at zero cost, the optimal value of problem (P_d) is bounded above by σ_θ^2 at every dimension. The processing cost paid by the optimal rule is therefore uniformly bounded. Assumption 2 then rules out any sequence of optimal estimates whose responsiveness remains bounded away from zero. Once the variance of the optimal estimate vanishes, the squared-error bound in (1) forces its mean to approach μ_θ .

No parametric signal structure is required for this conclusion. The signals may become increasingly precise or numerous, and the Bayesian posterior based on the full signal vector may remain highly informative. Proposition 2 concerns the estimate the agent optimally implements after accounting for processing costs. The statistical environment remains important for the value of learning at finite dimensions and for whether Assumption 2 is economically appropriate in a particular application.

Prior reversion gives a simple characterization of information overload.

Proposition 3. *Suppose Assumptions 1, 2, and 3 hold. Information overload occurs if and only if there exists a finite $d_0 \geq 0$ such that*

$$\text{MSE}(d_0) < \sigma_\theta^2. \quad (4)$$

Condition (4) is a finite-dimensional learning gain. It requires that, after the agent optimally trades off accuracy and processing cost, the resulting estimate be strictly more accurate than the prior mean at some finite dimension. The condition is therefore stronger than statistical informativeness alone. Signals may contain information about the state while processing costs make the prior-mean rule optimal at every dimension.

If no finite-dimensional environment improves on the prior benchmark, there is no accuracy gain that can subsequently be lost, and information overload does not occur. If a strict gain exists at some finite dimension, Proposition 2 implies that optimal accuracy eventually returns arbitrarily close to the prior benchmark. A finite dimension must then strictly outperform every richer environment.

Completely uninformative signals illustrate the first case. Suppose

$$\mathbb{E}[\theta \mid S^{(d)}] = \mu_\theta$$

at every dimension. For any $X = \ell(S^{(d)}) \in \mathcal{H}_d$, the conditional-expectation property gives

$$\mathbb{E}[(X - \theta)^2] = \sigma_\theta^2 + \mathbb{E}[(X - \mu_\theta)^2] \geq \sigma_\theta^2.$$

Because processing costs are nonnegative and the prior-mean rule attains σ_θ^2 at zero cost,

$$\text{MSE}(d) = \sigma_\theta^2$$

for every d . Such an environment produces no finite-dimensional learning gain and hence no information overload.

Information overload instead arises when responsive inference is valuable at a manageable dimension but becomes increasingly unattractive as the information environment expands. Propositions 2 and 3 characterize this limiting mechanism without determining the functional form of the optimal rule or the shape of the accuracy path at intermediate dimensions. The next section imposes additional structure on the posterior mean and processing technology to characterize the form of optimal inference.

4 The Structure of Optimal Inference

The preceding section characterized the limiting behavior of optimal inference while allowing the agent to choose any measurable function of the observed signals. We now identify conditions under which this infinite-dimensional problem reduces to a finite-dimensional choice of coefficients. The key question is whether nonlinear transformations of the signal vector can improve the agent's objective once both statistical accuracy and the cost of responsiveness are taken into account.

Under squared-error loss, the Bayesian posterior mean contains all of the signal-dependent variation that is useful for predicting the state. If that posterior mean is affine in the primitive signals, any nonaffine component of an estimate is orthogonal to the statistically relevant affine component. When processing costs are also weakly increasing in the variance of the induced estimate, such a nonaffine component raises both prediction error and responsiveness. It is therefore never optimal.

4.1 Affine Optimality

For the remainder of this section, suppose that

$$\mathbb{E}[\|S^{(d)}\|^2] < \infty$$

for every $d \geq 0$. Define the subspace of estimates that are affine in the primitive signals by

$$\mathcal{A}^{(d)} := \{c + \omega^\top S^{(d)} : \omega \in \mathbb{R}^{d+1}, c \in \mathbb{R}\} \subseteq \mathcal{H}_d.$$

Because $\mathcal{A}^{(d)}$ is the linear span of the finite collection

$$1, S_0, \dots, S_d,$$

it is finite-dimensional and therefore closed in $\mathcal{H}_d$.

Let

$$m_d := \mathbb{E}[\theta \mid S^{(d)}]$$

denote the Bayesian posterior mean. Under quadratic loss and in the absence of processing costs, m_d is the unique optimal estimate of θ in $\mathcal{H}_d$.

Assumption 4. *For every $d \geq 0$,*

$$m_d = \mathbb{E}[\theta \mid S^{(d)}] \in \mathcal{A}^{(d)}.$$

Assumption 4 requires the statistically optimal predictor to be affine in the primitive signal coordinates. It is satisfied when $(\theta, S^{(d)})$ is jointly Gaussian and, more generally, in environments with a linear conditional expectation. The assumption does not restrict the learning rule chosen by the agent; it restricts only the form of the frictionless Bayesian benchmark.

Our second restriction concerns the processing technology. At a fixed information dimension, estimates that vary more across signal realizations are weakly more costly to implement.

Assumption 5. *For every $d \geq 0$ and every $X, Y \in \mathcal{H}_d$,*

$$\text{Var}(X) \geq \text{Var}(Y) \implies C_d(X) \geq C_d(Y).$$

Thus, within a fixed information environment, responsiveness is sufficient for ranking processing costs. The cost may still depend separately on d , so the same degree of responsiveness can have different implementation costs in environments of different dimensions.

Proposition 4. *Suppose Assumptions 3, 4, and 5 hold. Then, for every $d \geq 0$, the unique optimal induced estimate belongs to $\mathcal{A}^{(d)}$. Equivalently, there exist*

$$\omega_d^* \in \mathbb{R}^{d+1} \quad \text{and} \quad c_d^* \in \mathbb{R}$$

such that

$$\ell_d^*(S^{(d)}) = (\omega_d^*)^\top S^{(d)} + c_d^*.$$

The induced estimate is unique up to almost-sure equality.

The proposition does not impose affine behavior as a restriction on the agent's strategy set. It derives affine optimality from the interaction between the statistical environment and the processing technology. To see the force of the result, take any candidate estimate and project it onto $\mathcal{A}^{(d)}$. Because the posterior mean belongs to $\mathcal{A}^{(d)}$, the nonaffine residual contains no component that improves prediction of the state. Removing it strictly lowers mean squared error whenever the residual is nonzero. The projection also weakly lowers the variance of the estimate and therefore cannot increase its processing cost. Every genuinely nonaffine estimate is consequently dominated.

Proposition 4 converts the original optimization over all measurable learning rules into a finite-dimensional coefficient problem.

Corollary 1. *Suppose Assumptions 3, 4, and 5 hold. Define*

$$\tilde{C}_d(\omega, c) := C_d(c + \omega^\top S^{(d)}).$$

Then problem (P_d) is equivalent to the finite-dimensional convex program

$$\min_{(\omega, c) \in \mathbb{R}^{d+1} \times \mathbb{R}} \left\{ \mathbb{E} \left[(c + \omega^\top S^{(d)} - \theta)^2 \right] + \tilde{C}_d(\omega, c) \right\}.$$

Every minimizing coefficient pair induces the unique optimal estimate $\ell_d^(S^{(d)})$.*

The coefficient representation need not itself be unique if the collection $1, S_0, \dots, S_d$ is linearly dependent in L^2 . Corollary 1 nevertheless guarantees that all minimizing coefficient pairs generate the same random estimate almost surely.

Affine optimality determines the functional form of the optimal rule at each fixed dimension, but it does not determine the coefficients or the path of accuracy as the information environment expands. We next impose a Gaussian signal structure and a quadratic responsiveness cost. This benchmark yields closed-form coefficients and makes the tradeoff between statistical precision and processing costs explicit.

5 Gaussian–Quadratic Benchmark

We now specialize the statistical environment and the processing technology. The Gaussian–quadratic benchmark serves two purposes. First, it gives a closed-form representation of the optimal estimate as a shrinkage transformation of the Bayesian posterior mean. Second, it permits a sharp distinction between two economic questions. When the information dimension is exogenous, we study the accuracy of the optimally chosen rule. When the dimension can itself be selected, we study the number of signals that minimizes total loss, including processing costs.

5.1 The Benchmark and the Cognitive Deflator

Fix $d \geq 0$ and write

$$m := d + 1$$

for the number of available signals. We specialize the processing cost to

$$C(\ell; d) := \lambda m \operatorname{Var}(\ell(S^{(d)})), \quad \lambda \geq 0. \quad (5)$$

The parameter λ converts responsiveness into processing cost, while the factor m captures the scale of the surrounding information environment. If an induced estimate has variance v , its processing cost is $\lambda m v$. Thus, holding the induced estimate fixed, adding one signal raises its implementation cost by λv .

The specification in (5) satisfies Assumptions 1, 3, and 5. When $\lambda > 0$, it also satisfies Assumption 2: any sequence of estimates whose responsiveness remains bounded away from zero has a processing cost that diverges with m . When $\lambda = 0$, the model reduces to the frictionless Bayesian benchmark. The linear dependence on m is imposed to obtain a transparent closed form, not as a universal description of processing complexity.

We impose the following statistical environment.

Assumption 6. *The state satisfies*

$$\theta \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2), \quad \sigma_\theta^2 > 0,$$

and the signals satisfy

$$S_k = \theta + \varepsilon_k, \quad k \geq 0,$$

where $(\varepsilon_k)_{k \geq 0}$ are independent and identically distributed according to

$$\varepsilon_k \sim \mathcal{N}(0, \sigma_\varepsilon^2), \quad \sigma_\varepsilon^2 > 0,$$

and are independent of θ .

Assumption 6 makes the sample average a sufficient statistic for the state and ensures that the posterior mean is affine. Define

$$\bar{S}^{(d)} := \frac{1}{m} \sum_{k=0}^d S_k.$$

The Bayesian posterior mean is

$$\mathbb{E}[\theta \mid S^{(d)}] = \mu_\theta + \beta_m (\bar{S}^{(d)} - \mu_\theta), \quad \beta_m := \frac{\sigma_\theta^2 m}{\sigma_\varepsilon^2 + \sigma_\theta^2 m}.$$

The coefficient $\beta_m \in (0, 1)$ is the Bayesian response to the sample average. It increases toward one as m grows because averaging additional independent signals reduces measurement noise.

Proposition 5. *Under Assumption 6 and the processing cost in (5), the unique optimal induced estimate is*

$$\ell_d^*(S^{(d)}) = \mu_\theta + \kappa_m (\bar{S}^{(d)} - \mu_\theta), \quad (6)$$

where

$$\kappa_m = \underbrace{\frac{1}{1 + \lambda m}}_{\text{cognitive deflator}} \underbrace{\frac{\sigma_\theta^2 m}{\sigma_\varepsilon^2 + \sigma_\theta^2 m}}_{\text{Bayesian responsiveness}} = \frac{1}{1 + \lambda m} \beta_m. \quad (7)$$

Equivalently,

$$\ell_d^*(S^{(d)}) = (\omega_d^*)^\top S^{(d)} + c_d^*,$$

where

$$\omega_d^* = \alpha_m \mathbf{1}_m, \quad \alpha_m = \frac{1}{1 + \lambda m} \frac{\sigma_\theta^2}{\sigma_\varepsilon^2 + \sigma_\theta^2 m}, \quad c_d^* = (1 - m\alpha_m) \mu_\theta.$$

The optimal estimate can be written directly as a deflated posterior mean:

$$\ell_d^*(S^{(d)}) = \mu_\theta + \frac{1}{1 + \lambda m} (\mathbb{E}[\theta \mid S^{(d)}] - \mu_\theta).$$

Processing costs therefore do not change the direction in which the observed signals move the estimate. They reduce the magnitude of the posterior-mean adjustment toward or away from the prior mean.

Equation (7) separates two effects of a larger information environment. The Bayesian responsiveness β_m increases with m , reflecting greater statistical precision. The cognitive deflator $(1 + \lambda m)^{-1}$ decreases with m , reflecting the higher cost of maintaining a responsive estimate. When $\lambda = 0$, the optimal rule is the Bayesian posterior mean. When $\lambda > 0$, Bayesian responsiveness converges to one, but the cognitive deflator converges to zero. The overall response κ_m therefore eventually vanishes.

5.2 Accuracy and the Preferred Information Dimension

We first treat d as exogenous and characterize the statistical accuracy of the optimal rule. From (6),

$$\ell_d^*(S^{(d)}) - \mu_\theta = \kappa_m \left(\theta - \mu_\theta + \frac{1}{m} \sum_{k=0}^d \varepsilon_k \right).$$

It follows that

$$\text{Var}(\ell_d^*(S^{(d)})) = \kappa_m^2 \left(\sigma_\theta^2 + \frac{\sigma_\varepsilon^2}{m} \right) = \frac{\sigma_\theta^4 m}{(\sigma_\varepsilon^2 + \sigma_\theta^2 m)(1 + \lambda m)^2}. \quad (8)$$

The corresponding mean squared error is

$$\text{MSE}(d) = (1 - \kappa_m)^2 \sigma_\theta^2 + \kappa_m^2 \frac{\sigma_\varepsilon^2}{m} \quad (9)$$

$$= \sigma_\theta^2 - \frac{\sigma_\theta^4 m(1 + 2\lambda m)}{(\sigma_\varepsilon^2 + \sigma_\theta^2 m)(1 + \lambda m)^2}. \quad (10)$$

The first term in (9) is the error generated by anchoring the estimate toward the prior mean. The second is the residual measurement error transmitted through the sample average. Averaging reduces the raw noise variance σ_ε^2/m , but the endogenous response κ_m is itself attenuated by the processing cost. The path of accuracy is determined by the interaction of these two forces.

Proposition 6. *Suppose Assumption 6 and the processing cost in (5) hold.*

(i) *If $\lambda = 0$, then*

$$\text{MSE}(d) = \frac{\sigma_\theta^2 \sigma_\varepsilon^2}{\sigma_\varepsilon^2 + \sigma_\theta^2(d + 1)},$$

so $\text{MSE}(d)$ is strictly decreasing in d and converges to zero.

(ii) *If $\lambda > 0$, define the continuous extension*

$$\mathcal{M}(m) := \sigma_\theta^2 - \frac{\sigma_\theta^4 m(1 + 2\lambda m)}{(\sigma_\varepsilon^2 + \sigma_\theta^2 m)(1 + \lambda m)^2}, \quad m > 0.$$

There exists a unique $\hat{m}_{\text{MSE}} > 0$ such that $\mathcal{M}$ is strictly decreasing on $(0, \hat{m}_{\text{MSE}})$ and strictly increasing on $(\hat{m}_{\text{MSE}}, \infty)$. Consequently, the discrete sequence $d \mapsto \text{MSE}(d) = \mathcal{M}(d+1)$ is single-troughed, and its set of minimizers consists of either one integer or two adjacent integers.

Moreover,

$$\text{MSE}(d) < \sigma_\theta^2 \quad \text{for every finite } d, \quad \lim_{d \rightarrow \infty} \text{MSE}(d) = \sigma_\theta^2.$$

Information overload therefore occurs whenever $\lambda > 0$.

For every finite d , the optimal responsive estimate is strictly more accurate than the prior mean. The size of this improvement nevertheless converges to zero. Depending on the parameters, accuracy may initially improve before deteriorating, or the feasible sequence may begin on the increasing side of the continuous MSE curve. In either case, some finite dimension is strictly more accurate than every sufficiently large environment.

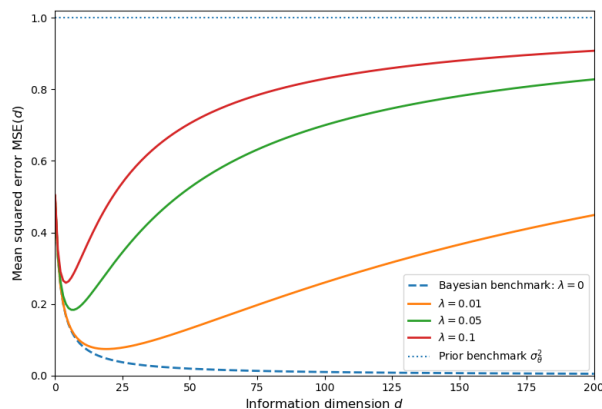


Figure 1: Information overload in the Gaussian–quadratic benchmark

The limiting behavior in Proposition 6 is an explicit instance of Proposition 2. When $\lambda > 0$,

$$\kappa_m \rightarrow 0, \quad \ell_d^*(S^{(d)}) \rightarrow \mu_\theta \quad \text{in } L^2, \quad \text{MSE}(d) \rightarrow \sigma_\theta^2.$$

When $\lambda = 0$, by contrast, $\kappa_m = \beta_m \rightarrow 1$, and averaging increasingly many independent signals permits arbitrarily accurate learning.

The preceding analysis compares statistical accuracy across exogenously given information dimensions. Suppose instead that the agent may choose d . Because the rule at each dimension already minimizes mean squared error plus processing cost conditional on d , the

relevant outer objective is the minimized total loss

$$L(d) := \text{MSE}(d) + \lambda m \text{Var}(\ell_d^*(S^{(d)})). \quad (11)$$

Substituting (8) and (10) gives

$$L(d) = \sigma_\theta^2 - \frac{\sigma_\theta^4 m}{(\sigma_\varepsilon^2 + \sigma_\theta^2 m)(1 + \lambda m)}, \quad m = d + 1. \quad (12)$$

Thus,

$$L(d) - \text{MSE}(d) = \lambda m \text{Var}(\ell_d^*(S^{(d)})) \geq 0.$$

Information overload is defined by the statistical accuracy $\text{MSE}(d)$. The preferred information dimension is instead determined by $L(d)$, which internalizes the resources used to implement the optimal estimate. The dimension minimizing MSE therefore need not coincide with the dimension minimizing total loss.

Proposition 7. *Suppose Assumption 6 and the processing cost in (5) hold.*

- (i) *If $\lambda = 0$, then $L(d)$ is strictly decreasing in d .*
- (ii) *If $\lambda > 0$, then, for $m = d + 1$,*

$$L(d + 1) - L(d) = \frac{\sigma_\theta^4 [\lambda \sigma_\theta^2 m(m + 1) - \sigma_\varepsilon^2]}{(\sigma_\varepsilon^2 + \sigma_\theta^2 m)(1 + \lambda m)(\sigma_\varepsilon^2 + \sigma_\theta^2(m + 1))(1 + \lambda(m + 1))}. \quad (13)$$

Consequently, $L(d)$ is single-troughed.

Define

$$x_\lambda := \frac{\sqrt{1 + 4\sigma_\varepsilon^2/(\lambda\sigma_\theta^2)} - 1}{2}. \quad (14)$$

If x_λ is not a positive integer, the unique minimizer is⁶

$$d_L^* = \lceil x_\lambda \rceil - 1.$$

If $x_\lambda = n$ for some positive integer n , the two minimizers are

$$d = n - 1 \quad \text{and} \quad d = n.$$

⁶ $\lceil x \rceil$ denotes the smallest integer greater than or equal to x .

In particular, $d = 0$ is the unique minimizer if

$$\lambda > \frac{\sigma_\varepsilon^2}{2\sigma_\theta^2},$$

while $d = 0$ and $d = 1$ are tied if equality holds.

Proposition 7 gives the exact solution to the discrete dimension problem. Recall that $d = 0$ corresponds to one observed signal. When processing costs are sufficiently high, the agent therefore chooses the smallest feasible information environment. At lower processing costs, averaging additional observations is worthwhile until the marginal statistical benefit is offset by the higher cost of maintaining responsiveness.

For $\lambda > 0$, the continuous relaxation over the feasible domain $m \geq 1$ is minimized at

$$m_L = \max \left\{ 1, \sqrt{\frac{\sigma_\varepsilon^2}{\lambda \sigma_\theta^2}} \right\}. \quad (15)$$

A higher signal-noise variance σ_ε^2 weakly increases the preferred number of observations because averaging becomes more valuable. A larger processing-cost coefficient λ reduces the preferred dimension. A higher prior variance σ_θ^2 also reduces the preferred dimension in this benchmark because a responsive estimate then exhibits greater variation and is consequently more expensive under the variance-based processing technology.

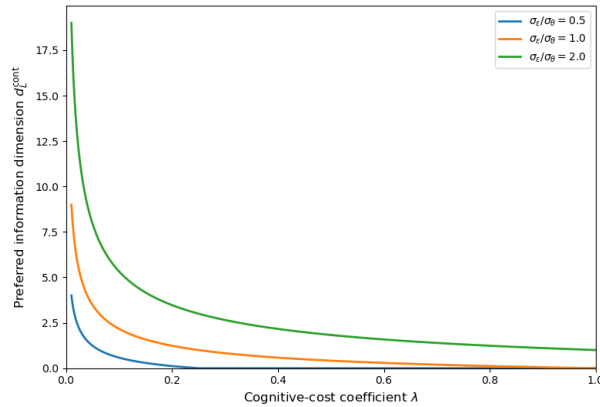


Figure 2: Preferred information dimension versus processing cost

5.3 Heterogeneous Signal Quality

The symmetric benchmark assigns equal weights to all signals because they have identical noise variances. We now allow signal quality to differ across sources. The extension

shows that the processing cost governs the overall scale of responsiveness, while the relative allocation of weight across signals continues to reflect their statistical precision.

Assumption 7. *The state satisfies*

$$\theta \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2),$$

and the signals satisfy

$$S_k = \theta + \varepsilon_k, \quad k \geq 0,$$

where the noise terms are mutually independent, independent of θ , and satisfy

$$\varepsilon_k \sim \mathcal{N}(0, \sigma_{\varepsilon,k}^2), \quad \sigma_{\varepsilon,k}^2 > 0.$$

For a fixed dimension d , define total signal precision by

$$P_d := \sum_{k=0}^d \frac{1}{\sigma_{\varepsilon,k}^2}.$$

Proposition 8. *Under Assumption 7 and the processing cost in (5), the unique optimal rule is*

$$\ell_d^*(S^{(d)}) = (\omega_d^*)^\top S^{(d)} + c_d^*, \quad c_d^* = (1 - (\omega_d^*)^\top \mathbf{1}_m) \mu_\theta,$$

where

$$\omega_{d,k}^* = \underbrace{\frac{1}{1 + \lambda m}}_{\text{cognitive deflator}} \underbrace{\frac{\sigma_\theta^2}{1 + \sigma_\theta^2 P_d} \frac{1}{\sigma_{\varepsilon,k}^2}}_{\text{Bayesian weight}}, \quad k = 0, 1, \dots, d. \quad (16)$$

Consequently, for any $j, k \in \{0, \dots, d\}$,

$$\sigma_{\varepsilon,k}^2 < \sigma_{\varepsilon,j}^2 \iff \omega_{d,k}^* > \omega_{d,j}^*,$$

and

$$\frac{\omega_{d,k}^*}{\omega_{d,j}^*} = \frac{\sigma_{\varepsilon,j}^2}{\sigma_{\varepsilon,k}^2}.$$

As in the symmetric benchmark, the optimal estimate is a deflated version of the Bayesian posterior mean:

$$\ell_d^*(S^{(d)}) = \mu_\theta + \frac{1}{1 + \lambda m} (\mathbb{E}[\theta \mid S^{(d)}] - \mu_\theta).$$

The common factor $(1 + \lambda m)^{-1}$ reduces the overall response to the signal vector. It does not distort the relative use of the different signals. More precise signals receive greater weight,

and the ratios of the optimal coefficients coincide with the corresponding ratios of Bayesian coefficients.

The exact dimension comparative statics in Propositions 6 and 7 rely on signals of equal quality. With heterogeneous noise variances, information dimension alone no longer summarizes the statistical gain from expanding the signal vector. Adding a highly precise signal can be substantially more valuable than adding a nearly uninformative one. The heterogeneous extension therefore preserves the common-deflator and relative-weighting conclusions, while the paths of MSE and total loss depend on the sequence $\{\sigma_{\varepsilon,k}^2\}_{k \geq 0}$.

6 Processing Costs and Correlation Neglect

Correlation neglect refers to treating information from correlated sources as though the sources were independent (Dasaratha and He, 2021; Enke and Zimmermann, 2019). When several signals contain a common noise component, this misspecification leads the decision maker to overstate the amount of independent evidence and respond too strongly to the signals.

We take the misspecified covariance model as given and study how dimension-dependent processing costs affect its behavioral and payoff consequences. The comparison is between two inference rules. A correlation-aware agent chooses the optimal rule under the true covariance structure, while a correlation-neglect agent chooses the optimal rule under a subjective model in which the signals are independent. Both agents face the same functional form for processing costs within their respective models.

The central observation is that processing costs act on the absolute responsiveness of an estimate. In the Gaussian–quadratic benchmark, they introduce the same dimension-dependent deflator into both rules. Correlation neglect continues to generate excessive responsiveness relative to the correctly specified benchmark, but the absolute response of both agents may become small. The two agents can therefore behave similarly in large information environments even though their covariance beliefs remain different.

6.1 Correlated Signals and Inference Rules

Fix $d \geq 0$ and write $m := d + 1$. Let

$$\theta \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2), \quad S_k = \theta + \varepsilon_k, \quad k \geq 0.$$

The noise sequence is jointly Gaussian, centered, and independent of θ , with⁷

$$\text{Var}(\varepsilon_k) = \sigma_\varepsilon^2, \quad \text{Cov}(\varepsilon_j, \varepsilon_k) = \rho\sigma_\varepsilon^2 \quad \text{for } j \neq k.$$

The parameter ρ measures the common component of signal noise. When $\rho = 0$, the signal noises are independent. When $\rho > 0$, averaging eliminates the idiosyncratic components but not the common component.

Let

$$\bar{S}^{(d)} := \frac{1}{m} \sum_{k=0}^d S_k$$

denote the sample average and define

$$q_m := \rho + \frac{1 - \rho}{m}. \tag{17}$$

Then

$$\text{Var}(\bar{S}^{(d)} - \theta) = \sigma_\varepsilon^2 q_m, \quad \text{Var}(\bar{S}^{(d)}) = \sigma_\theta^2 + \sigma_\varepsilon^2 q_m.$$

If $\rho = 0$, then $q_m = 1/m$, and the noise variance of the average converges to zero. If $\rho > 0$, then $q_m \rightarrow \rho$, so the average retains noise variance $\rho\sigma_\varepsilon^2$ regardless of how many signals are observed.

We maintain the quadratic processing cost

$$C(\ell; d) = \lambda m \text{Var}(\ell(S^{(d)})), \quad \lambda \geq 0.$$

Under the true covariance structure, the sample average is sufficient for θ , and the Bayesian posterior mean is

$$\mathbb{E}[\theta \mid S^{(d)}] = \mu_\theta + \beta_m^A (\bar{S}^{(d)} - \mu_\theta), \quad \beta_m^A := \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_\varepsilon^2 q_m}.$$

The superscript A denotes correlation awareness. Applying the same quadratic optimization

⁷One construction is obtained by taking $U, U_0, U_1, \dots$ to be independent standard normal random variables, independent of θ , and defining

$$\varepsilon_k = \sigma_\varepsilon (\sqrt{\rho} U + \sqrt{1 - \rho} U_k), \quad \rho \in [0, 1).$$

as in Proposition 5, the correlation-aware optimal rule is

$$\ell_d^A(S^{(d)}) = \mu_\theta + \kappa_m^A \left(\bar{S}^{(d)} - \mu_\theta \right), \quad \kappa_m^A := \frac{1}{1 + \lambda m} \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_\varepsilon^2 q_m}. \quad (18)$$

Equivalently,

$$\omega_m^A = \frac{\kappa_m^A}{m} \mathbf{1}_m.$$

A higher ρ makes the sample average less informative and therefore lowers the correlation-aware response.

Now consider an agent who believes that the signal noises are independent. For the purpose of choosing her inference rule, she sets $\rho = 0$ when evaluating both statistical error and the variance entering her processing cost. She therefore solves the independent-signal problem from Section 5. The resulting correlation-neglect rule is

$$\ell_d^N(S^{(d)}) = \mu_\theta + \kappa_m^N \left(\bar{S}^{(d)} - \mu_\theta \right), \quad \kappa_m^N := \frac{1}{1 + \lambda m} \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_\varepsilon^2/m}, \quad (19)$$

with coefficient vector

$$\omega_m^N = \frac{\kappa_m^N}{m} \mathbf{1}_m.$$

The superscript N denotes correlation neglect. The rule ℓ_d^N is optimal under the subjective independent-noise model. Its realized behavior and performance below are evaluated under the true correlated-signal distribution.

For $\rho > 0$, the correlation-neglect rule assigns excessive weight to the sample average. Define the response gap by

$$\Delta_m := \kappa_m^N - \kappa_m^A = \frac{1}{1 + \lambda m} \frac{\sigma_\theta^2 \sigma_\varepsilon^2 \rho (m - 1)/m}{(\sigma_\theta^2 + \sigma_\varepsilon^2/m) (\sigma_\theta^2 + \sigma_\varepsilon^2 q_m)}. \quad (20)$$

Hence,

$$\Delta_m \geq 0,$$

with strict inequality whenever $\rho > 0$ and $m > 1$. The two rules coincide when only one signal is observed because correlation across sources is then irrelevant. Once multiple sources are present, correlation neglect generates overreaction by treating partly redundant observations as independent evidence.

Both rules nevertheless contain the common cognitive deflator

$$\frac{1}{1 + \lambda m}.$$

Correlation neglect changes the Bayesian component of the response, while the processing technology determines how strongly either Bayesian component is expressed in the final estimate. The next subsection shows that this distinction determines whether the behavioral and payoff effects of misspecification persist in large information environments.

6.2 Behavioral and Payoff Consequences

Define the true total-loss functional at information size m by

$$\mathcal{J}_m(\ell) := \mathbb{E} \left[\left(\ell(S^{(d)}) - \theta \right)^2 \right] + \lambda m \text{Var}(\ell(S^{(d)})), \quad (21)$$

where expectations and variances are evaluated under the true correlated-signal distribution. By construction, ℓ_d^A minimizes $\mathcal{J}_m$. The rule ℓ_d^N instead minimizes the corresponding subjective objective under the independent-noise model and need not minimize true total loss.

Proposition 9. *Suppose $\rho \in (0, 1)$.⁸*

(i) *If $\lambda > 0$, then*

$$\ell_d^A(S^{(d)}) \rightarrow \mu_\theta \quad \text{and} \quad \ell_d^N(S^{(d)}) \rightarrow \mu_\theta \quad \text{in } L^2$$

as $d \rightarrow \infty$. Moreover, writing $m = d + 1$,

$$\|\ell_d^N(S^{(d)}) - \ell_d^A(S^{(d)})\|_2 = O\left(\frac{1}{m}\right), \quad (22)$$

and

$$\|\omega_m^N - \omega_m^A\|_E = O\left(\frac{1}{m^{3/2}}\right), \quad (23)$$

where $\|\cdot\|_E$ denotes the Euclidean norm on coefficient vectors. The difference in true total loss satisfies

$$0 \leq \mathcal{J}_m(\ell_d^N) - \mathcal{J}_m(\ell_d^A) = (1 + \lambda m) (\sigma_\theta^2 + \sigma_\varepsilon^2 q_m) \Delta_m^2 = O\left(\frac{1}{m}\right). \quad (24)$$

(ii) *If $\lambda = 0$, then*

$$\kappa_m^A \longrightarrow \frac{\sigma_\theta^2}{\sigma_\theta^2 + \rho \sigma_\varepsilon^2}, \quad \kappa_m^N \longrightarrow 1.$$

⁸We use standard asymptotic notation: $x_m = O(y_m)$ means that there exist constants $K < \infty$ and M such that $|x_m| \leq K|y_m|$ for every $m \geq M$.

Consequently,

$$\lim_{d \rightarrow \infty} \mathbb{E} \left[\left(\ell_d^N(S^{(d)}) - \ell_d^A(S^{(d)}) \right)^2 \right] = \frac{\rho^2 \sigma_\varepsilon^4}{\sigma_\theta^2 + \rho \sigma_\varepsilon^2} > 0, \quad (25)$$

and

$$\lim_{d \rightarrow \infty} [\mathcal{J}_m(\ell_d^N) - \mathcal{J}_m(\ell_d^A)] = \frac{\rho^2 \sigma_\varepsilon^4}{\sigma_\theta^2 + \rho \sigma_\varepsilon^2} > 0.$$

Part (i) establishes asymptotic behavioral similarity under positive processing costs. The Bayesian components of the two rules remain different: the correlation-aware rule discounts the common noise component, while the correlation-neglect rule treats the sample average as increasingly precise. The common cognitive deflator nevertheless makes the absolute response of both rules vanish at rate $1/m$. Consequently, both estimates converge to the prior mean, and their difference converges to zero.

Equation (22) concerns the distance between the realized estimates under the true distribution. It is therefore stronger economically than the coefficient comparison in (23). A small Euclidean distance between coefficient vectors does not by itself imply similar behavior when the number and covariance of the signals are changing. Here, the induced estimates themselves become close.

The exact identity in (24) gives the corresponding payoff implication. Because ℓ_d^A minimizes the true objective, correcting the covariance misspecification can only reduce true total loss. With $\lambda > 0$, however, the value of that correction converges to zero. Processing costs do not make the subjective covariance model accurate. They make the chosen estimate increasingly insensitive to the component of the model on which the two agents disagree.

Part (ii) shows that dimensionality alone does not generate this convergence. When $\lambda = 0$, the correlation-neglect agent becomes fully responsive to the sample average:

$$\kappa_m^N \rightarrow 1.$$

The correlation-aware agent instead continues to discount the common noise:

$$\kappa_m^A \rightarrow \frac{\sigma_\theta^2}{\sigma_\theta^2 + \rho \sigma_\varepsilon^2} < 1.$$

Because the common component cannot be averaged away, the limiting difference between the two estimates remains nondegenerate, as does the true loss from using the misspecified rule.

The proposition has an identification implication. Similar behavior in a large information environment need not indicate that decision makers hold similar beliefs about the covariance

structure. A correlation-aware and a correlation-neglect agent can produce nearly identical estimates because both optimally respond only weakly to the available signals. Processing costs can therefore mask a persistent statistical misspecification in observed behavior and reduce its payoff consequences without correcting the underlying model.

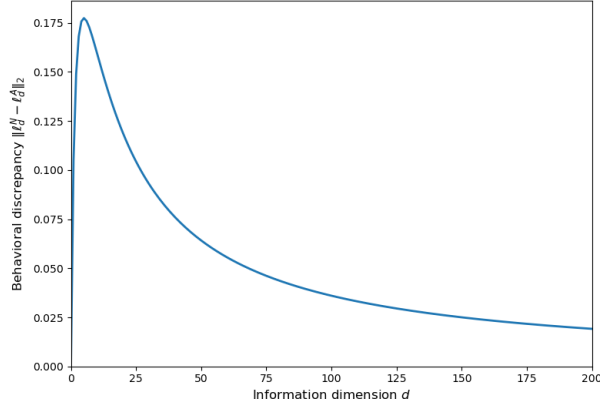


Figure 3: Behavioral gap between correlation-aware and correlation-neglect rules

7 Conclusion

This paper studies optimal inference when the cost of implementing a responsive estimate increases with the dimension of the information environment. The agent observes the full signal vector and chooses how to transform it into an estimate of an unknown state. Additional signals expand the set of statistically feasible estimates, but they may also raise the cost of maintaining a given degree of responsiveness. When constant estimates are costless and every sequence with nonvanishing responsiveness becomes prohibitively costly, the optimal estimate converges in mean square to the prior mean and its accuracy returns to the prior benchmark. Information overload occurs precisely when some finite information dimension initially generates a strict accuracy gain that is subsequently lost.

We also characterize the structure of optimal inference. When the Bayesian posterior mean is affine and processing costs are increasing in responsiveness, the optimal estimate is affine in the primitive signals. In the Gaussian–quadratic benchmark, it equals the Bayesian posterior mean shrunk toward the prior by a dimension-dependent cognitive deflator. A larger signal set then creates two opposing forces: additional observations improve statistical precision, while the rising cost of responsiveness strengthens shrinkage toward the prior. Their interaction generates a single-troughed mean squared error path and a finite preferred information dimension when processing costs are positive. With heterogeneous signal quality, the

common deflator reduces overall responsiveness while preserving Bayesian relative weights across signals.

Finally, dimension-dependent shrinkage changes the consequences of statistical misspecification. A correlation-neglect agent overreacts to redundant signals, but positive processing costs attenuate both the misspecified and correctly specified rules. In sufficiently large information environments, their estimates and true total losses become arbitrarily close even though their covariance models remain different. Without processing costs, the behavioral and payoff gaps persist. More information therefore need not produce more informative behavior: when implementing responsiveness becomes increasingly costly, a rational decision maker may optimally rely less on an expanding set of signals, and observable behavior may reveal less about the statistical model underlying her inference.

A Proofs

Proof of Proposition 1

Fix $d \in \mathbb{N}_0$ and work on the Hilbert space $\mathcal{H}_d$. By Assumption 3(i), there exists $X_0 \in \mathcal{H}_d$ with $C_d(X_0) < +\infty$. Since $X_0, \theta \in L^2$, this implies $J_d(X_0) < +\infty$, so J_d is proper.

Because $C_d \geq 0$,

$$J_d(X) \geq \mathbb{E}[(X - \theta)^2] = \|X - \theta\|_2^2 \geq (\|X\|_2 - \|\theta\|_2)^2.$$

Thus $J_d(X) \rightarrow +\infty$ as $\|X\|_2 \rightarrow +\infty$. Let $\{X_n\} \subseteq \mathcal{H}_d$ be a minimizing sequence. Coercivity implies that $\{X_n\}$ is bounded, so, because $\mathcal{H}_d$ is a Hilbert space, some subsequence converges weakly to an element $X^* \in \mathcal{H}_d$. The map $X \mapsto \|X - \theta\|_2^2$ is weakly lower semicontinuous. Moreover, a convex functional that is lower semicontinuous in the norm topology is weakly lower semicontinuous, so Assumption 3(ii)–(iii) implies that C_d is weakly lower semicontinuous. Therefore

$$J_d(X^*) \leq \liminf_{n \rightarrow \infty} J_d(X_n) = \inf_{X \in \mathcal{H}_d} J_d(X),$$

and X^* attains the minimum.

To prove uniqueness, suppose X_1 and X_2 are two minimizers that are not equal almost surely.

Their objective values, and hence their costs, are finite. For every $\alpha \in (0, 1)$,

$$\begin{aligned} \mathbb{E}\left[(\alpha X_1 + (1 - \alpha)X_2 - \theta)^2\right] &= \alpha \mathbb{E}[(X_1 - \theta)^2] + (1 - \alpha) \mathbb{E}[(X_2 - \theta)^2] \\ &\quad - \alpha(1 - \alpha) \|X_1 - X_2\|_2^2. \end{aligned}$$

By convexity of C_d ,

$$C_d(\alpha X_1 + (1 - \alpha)X_2) \leq \alpha C_d(X_1) + (1 - \alpha)C_d(X_2).$$

Since $\|X_1 - X_2\|_2^2 > 0$, adding the two displays gives

$$J_d(\alpha X_1 + (1 - \alpha)X_2) < \alpha J_d(X_1) + (1 - \alpha)J_d(X_2),$$

contradicting optimality of X_1 and X_2 . Hence the optimal induced estimate is unique up to almost-sure equality. ■

Proof of Proposition 2

Fix any selection of optimal rules and set $X_d := \ell_d^*(S^{(d)})$ and $v_d := \text{Var}(X_d)$. By Assumption 1, the prior-mean rule is feasible at zero cost, so optimality gives

$$\mathbb{E}[(X_d - \theta)^2] + C(\ell_d^*; d) \leq \sigma_\theta^2$$

for every $d \in \mathbb{N}_0$. In particular,

$$\mathbb{E}[(X_d - \theta)^2] \leq \sigma_\theta^2 \quad \text{and} \quad C(\ell_d^*; d) \leq \sigma_\theta^2.$$

We first prove that $v_d \rightarrow 0$. Suppose otherwise. Then there exist $\eta > 0$ and a strictly increasing sequence $\{d_k\}_{k \geq 1}$ such that $v_{d_k} \geq \eta$ for every k . For $d \geq d_1$, let $k(d) := \max\{k : d_k \leq d\}$ and define the lifted rule

$$\widehat{\ell}_d(s_0, \dots, s_d) := \ell_{d_{k(d)}}^*(s_0, \dots, s_{d_{k(d)}}).$$

For $d < d_1$, set $\widehat{\ell}_d \equiv \mu_\theta$. By nestedness,

$$\widehat{\ell}_d(S^{(d)}) = \ell_{d_{k(d)}}^*(S^{(d_{k(d)})})$$

almost surely for every $d \geq d_1$, and therefore

$$\text{Var}\left(\widehat{\ell}_d(S^{(d)})\right) = v_{d_{k(d)}} \geq \eta.$$

Hence $\liminf_{d \rightarrow \infty} \text{Var}(\widehat{\ell}_d(S^{(d)})) \geq \eta > 0$, so Assumption 2 requires $C(\widehat{\ell}_d; d) \rightarrow +\infty$. Along the subsequence $d = d_k$, however, $\widehat{\ell}_{d_k} = \ell_{d_k}^*$, and thus

$$C(\widehat{\ell}_{d_k}; d_k) = C(\ell_{d_k}^*; d_k) \leq \sigma_\theta^2$$

for every k , a contradiction. Therefore $v_d \rightarrow 0$.

Let $b_d := \mathbb{E}[X_d] - \mu_\theta$. Expanding the mean squared error and using the preceding upper bound gives

$$\sigma_\theta^2 + v_d + b_d^2 - 2 \text{Cov}(X_d, \theta) = \mathbb{E}[(X_d - \theta)^2] \leq \sigma_\theta^2.$$

Consequently,

$$0 \leq v_d + b_d^2 \leq 2 \text{Cov}(X_d, \theta) \leq 2 |\text{Cov}(X_d, \theta)| \leq 2\sigma_\theta \sqrt{v_d},$$

where the last inequality is Cauchy–Schwarz. Since $v_d \rightarrow 0$, it follows that $b_d \rightarrow 0$, and hence

$$\mathbb{E}[(X_d - \mu_\theta)^2] = v_d + b_d^2 \rightarrow 0.$$

This proves (2). The variance and mean conclusions follow from $v_d \rightarrow 0$ and $b_d \rightarrow 0$. Finally, the reverse triangle inequality gives

$$\left| \sqrt{\text{MSE}(d)} - \sigma_\theta \right| = \left| \|X_d - \theta\|_2 - \|\mu_\theta - \theta\|_2 \right| \leq \|X_d - \mu_\theta\|_2 \rightarrow 0,$$

so $\text{MSE}(d) \rightarrow \sigma_\theta^2$. ■

Proof of Proposition 3

Necessity. Suppose information overload occurs at d_1 . Then

$$\text{MSE}(d_1) < \text{MSE}(d_1 + 1) \leq \sigma_\theta^2,$$

where the second inequality follows from (1). Thus $\text{MSE}(d_1) < \sigma_\theta^2$, so (4) holds with $d_0 = d_1$.

Sufficiency. Suppose there exists d_0 such that $\text{MSE}(d_0) < \sigma_\theta^2$, and define

$$\delta := \frac{\sigma_\theta^2 - \text{MSE}(d_0)}{2} > 0.$$

By Proposition 2, there exists an integer $D \geq d_0$ such that

$$\text{MSE}(d) > \sigma_\theta^2 - \delta = \frac{\sigma_\theta^2 + \text{MSE}(d_0)}{2} > \text{MSE}(d_0)$$

for every $d > D$. Let d_1 be the largest minimizer of $\text{MSE}(d)$ over the finite set $\{0, 1, \dots, D\}$. If $d_1 < d \leq D$, maximality of d_1 implies $\text{MSE}(d) > \text{MSE}(d_1)$. If $d > D$, then

$$\text{MSE}(d) > \text{MSE}(d_0) \geq \text{MSE}(d_1),$$

where the weak inequality follows because $d_0 \leq D$ and d_1 is a minimizer on $\{0, \dots, D\}$. Therefore $\text{MSE}(d) > \text{MSE}(d_1)$ for every $d > d_1$, which is information overload. ■

Proof of Proposition 4

Fix $d \in \mathbb{N}_0$. Because $\mathcal{A}^{(d)}$ is a closed subspace of $\mathcal{H}_d$, the orthogonal projection $P_{\mathcal{A}^{(d)}} : \mathcal{H}_d \rightarrow \mathcal{A}^{(d)}$ is well defined. For any $X \in \mathcal{H}_d$, write

$$X = A + R, \quad A := P_{\mathcal{A}^{(d)}} X, \quad R := X - A \in (\mathcal{A}^{(d)})^\perp.$$

Let $M_d := \mathbb{E}[\theta \mid S^{(d)}]$ and $U_d := \theta - M_d$. Assumption 4 gives $M_d \in \mathcal{A}^{(d)}$, while $\mathbb{E}[U_d Y] = 0$ for every $Y \in \mathcal{H}_d$. Since $A - M_d \in \mathcal{A}^{(d)}$, $R \perp \mathcal{A}^{(d)}$, and $U_d \perp \mathcal{H}_d$, the Pythagorean identity yields

$$\mathbb{E}[(X - \theta)^2] = \|A - M_d\|_2^2 + \|R\|_2^2 + \|U_d\|_2^2 = \mathbb{E}[(A - \theta)^2] + \|R\|_2^2.$$

Thus replacing X by A weakly reduces mean squared error, and the reduction is strict whenever $R \neq 0$ in L^2 .

Because $1 \in \mathcal{A}^{(d)}$, orthogonality gives $\mathbb{E}[R] = 0$. Moreover, $A - \mathbb{E}[A] \in \mathcal{A}^{(d)}$, so $\text{Cov}(A, R) = 0$. Therefore

$$\text{Var}(X) = \text{Var}(A) + \text{Var}(R) = \text{Var}(A) + \|R\|_2^2 \geq \text{Var}(A).$$

Assumption 5 then implies $C_d(X) \geq C_d(A)$.

Let X_d^* be the unique minimizer from Proposition 1. Its objective value and cost are finite.

If $X_d^* \notin \mathcal{A}^{(d)}$, its projection $A_d^* := P_{\mathcal{A}^{(d)}} X_d^*$ satisfies $C_d(A_d^*) \leq C_d(X_d^*) < +\infty$ and

$$\mathbb{E}[(A_d^* - \theta)^2] < \mathbb{E}[(X_d^* - \theta)^2].$$

Hence $J_d(A_d^*) < J_d(X_d^*)$, contradicting optimality. Therefore $X_d^* \in \mathcal{A}^{(d)}$, so $X_d^* = c_d^* + (\omega_d^*)^\top S^{(d)}$ for some $c_d^* \in \mathbb{R}$ and $\omega_d^* \in \mathbb{R}^{d+1}$. The induced estimate is unique, although its coefficient representation need not be unique if $1, S_0, \dots, S_d$ are linearly dependent in L^2 . ■

Proof of Corollary 1

Fix $d \in \mathbb{N}_0$ and define the linear map

$$T_d : \mathbb{R}^{d+1} \times \mathbb{R} \rightarrow \mathcal{H}_d, \quad T_d(\omega, c) := c + \omega^\top S^{(d)}.$$

Its range is $\mathcal{A}^{(d)}$. By Proposition 4, the unique optimal induced estimate for problem (P_d) belongs to $\mathcal{A}^{(d)}$. Restricting the original problem to the range of T_d therefore does not change its minimum.

Because T_d is linear, the function

$$(\omega, c) \mapsto \mathbb{E}[(T_d(\omega, c) - \theta)^2]$$

is convex. Moreover,

$$\tilde{C}_d(\omega, c) = C_d(T_d(\omega, c))$$

is convex because C_d is convex and T_d is linear. Hence the coefficient problem is convex.

The optimal induced estimate has at least one coefficient representation because it belongs to $\mathcal{A}^{(d)}$, so the coefficient problem admits a minimizer. Every minimizing coefficient pair induces an optimal estimate for problem (P_d) ; uniqueness of the optimal induced estimate then implies that all minimizing coefficient pairs generate $\ell_d^*(S^{(d)})$ almost surely. ■

Proof of Proposition 5

Fix $d \in \mathbb{N}_0$ and write $m := d + 1$. Under Assumption 6,

$$S^{(d)} = \theta \mathbf{1}_m + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2 I_m),$$

where ε is independent of θ . By Proposition 4, it is enough to consider estimates of the form $\ell(S^{(d)}) = \omega^\top S^{(d)} + c$.

For fixed ω , the variance term does not depend on c , and the first-order condition for the squared-error term is

$$\mathbb{E}[\omega^\top S^{(d)} + c - \theta] = 0.$$

Since $\mathbb{E}[S^{(d)}] = \mu_\theta \mathbf{1}_m$, the optimal intercept is

$$c = (1 - \omega^\top \mathbf{1}_m) \mu_\theta.$$

Let $\tilde{\theta} := \theta - \mu_\theta$ and $\tilde{S} := S^{(d)} - \mu_\theta \mathbf{1}_m$. With the optimal intercept,

$$\ell(S^{(d)}) - \mu_\theta = \omega^\top \tilde{S}, \quad \ell(S^{(d)}) - \theta = \omega^\top \tilde{S} - \tilde{\theta}.$$

Define

$$\Sigma := \text{Var}(\tilde{S}) = \sigma_\theta^2 \mathbf{1}_m \mathbf{1}_m^\top + \sigma_\varepsilon^2 I_m, \quad \text{Cov}(\tilde{S}, \tilde{\theta}) = \sigma_\theta^2 \mathbf{1}_m.$$

The objective, after optimizing over c , is

$$\sigma_\theta^2 - 2\sigma_\theta^2 \omega^\top \mathbf{1}_m + (1 + \lambda m) \omega^\top \Sigma \omega.$$

Because Σ is positive definite, this is a strictly convex quadratic in ω . Its unique minimizer satisfies

$$(1 + \lambda m) \Sigma \omega = \sigma_\theta^2 \mathbf{1}_m,$$

and therefore

$$\omega = \frac{\sigma_\theta^2}{1 + \lambda m} \Sigma^{-1} \mathbf{1}_m.$$

The Sherman–Morrison formula⁹ gives

$$\Sigma^{-1} = \frac{1}{\sigma_\varepsilon^2} I_m - \frac{\sigma_\theta^2}{\sigma_\varepsilon^2(\sigma_\varepsilon^2 + \sigma_\theta^2 m)} \mathbf{1}_m \mathbf{1}_m^\top, \quad \Sigma^{-1} \mathbf{1}_m = \frac{1}{\sigma_\varepsilon^2 + \sigma_\theta^2 m} \mathbf{1}_m.$$

Hence

$$\omega_d^* = \alpha_m \mathbf{1}_m, \quad \alpha_m = \frac{1}{1 + \lambda m} \frac{\sigma_\theta^2}{\sigma_\varepsilon^2 + \sigma_\theta^2 m},$$

⁹If A is invertible and u, v are conformable vectors with $1 + v^\top A^{-1} u \neq 0$, then

$$(A + uv^\top)^{-1} = A^{-1} - \frac{A^{-1} u v^\top A^{-1}}{1 + v^\top A^{-1} u}.$$

Here, take $A = \sigma_\varepsilon^2 I_m$, $u = \sigma_\theta^2 \mathbf{1}_m$, and $v = \mathbf{1}_m$.

and

$$c_d^* = (1 - m\alpha_m)\mu_\theta.$$

Finally,

$$(\omega_d^*)^\top S^{(d)} + c_d^* = \mu_\theta + m\alpha_m(\bar{S}^{(d)} - \mu_\theta),$$

and

$$m\alpha_m = \frac{1}{1 + \lambda m} \frac{\sigma_\theta^2 m}{\sigma_\varepsilon^2 + \sigma_\theta^2 m} = \kappa_m.$$

This proves both representations in the proposition. ■

Proof of Proposition 6

If $\lambda = 0$, equation (10) gives

$$\text{MSE}(d) = \sigma_\theta^2 - \frac{\sigma_\theta^4 m}{\sigma_\varepsilon^2 + \sigma_\theta^2 m} = \frac{\sigma_\theta^2 \sigma_\varepsilon^2}{\sigma_\varepsilon^2 + \sigma_\theta^2 m}, \quad m = d + 1.$$

This expression is strictly decreasing in m and converges to zero, proving part (i).

Now suppose $\lambda > 0$. Differentiating the continuous extension in the proposition yields

$$\mathcal{M}'(m) = \frac{\sigma_\theta^4 (2\sigma_\theta^2 \lambda^2 m^3 - 3\sigma_\varepsilon^2 \lambda m - \sigma_\varepsilon^2)}{(\sigma_\varepsilon^2 + \sigma_\theta^2 m)^2 (1 + \lambda m)^3}.$$

The denominator is strictly positive. Define

$$h(m) := 2\sigma_\theta^2 \lambda^2 m^3 - 3\sigma_\varepsilon^2 \lambda m - \sigma_\varepsilon^2.$$

Then $h(0) = -\sigma_\varepsilon^2 < 0$, $h(m) \rightarrow +\infty$, and

$$h'(m) = 6\sigma_\theta^2 \lambda^2 m^2 - 3\sigma_\varepsilon^2 \lambda.$$

Thus h is strictly decreasing on $(0, m_0)$ and strictly increasing on (m_0, ∞) , where

$$m_0 := \sqrt{\frac{\sigma_\varepsilon^2}{2\sigma_\theta^2 \lambda}}.$$

Because $h(m_0) < h(0) < 0$, the function h crosses zero exactly once on $(0, \infty)$. Denote this root by $\hat{m}_{\text{MSE}}$. The sign of $\mathcal{M}'(m)$ is therefore negative for $m < \hat{m}_{\text{MSE}}$ and positive for $m > \hat{m}_{\text{MSE}}$, proving the asserted single-trough shape.

If $\widehat{m}_{\text{MSE}} \leq 1$, then $\mathcal{M}$ is strictly increasing on $[1, \infty)$, so $m = 1$ is the unique integer minimizer. If $\widehat{m}_{\text{MSE}} > 1$, strict monotonicity implies that every integer below $\lfloor \widehat{m}_{\text{MSE}} \rfloor$ is dominated by $\lfloor \widehat{m}_{\text{MSE}} \rfloor$, while every integer above $\lceil \widehat{m}_{\text{MSE}} \rceil$ is dominated by $\lceil \widehat{m}_{\text{MSE}} \rceil$. The integer minimizers therefore lie in the one- or two-element set

$$\{\lfloor \widehat{m}_{\text{MSE}} \rfloor, \lceil \widehat{m}_{\text{MSE}} \rceil\} \cap \mathbb{N},$$

so there is either one minimizing integer or two adjacent minimizing integers. Since $m = d+1$, the same conclusion holds for $d \mapsto \text{MSE}(d)$.

For every finite $m > 0$,

$$\mathcal{M}(m) = \sigma_\theta^2 - \frac{\sigma_\theta^4 m (1 + 2\lambda m)}{(\sigma_\varepsilon^2 + \sigma_\theta^2 m) (1 + \lambda m)^2} < \sigma_\theta^2,$$

because the subtracted term is strictly positive. Moreover, this term is $O(m^{-1})$, so $\mathcal{M}(m) \rightarrow \sigma_\theta^2$. Hence $\text{MSE}(d) < \sigma_\theta^2$ for every finite d and $\text{MSE}(d) \rightarrow \sigma_\theta^2$. Information overload follows from Proposition 3. ■

Proof of Proposition 7

Write $m := d + 1$. By (12),

$$L(d) = \sigma_\theta^2 - \frac{\sigma_\theta^4 m}{(\sigma_\varepsilon^2 + \sigma_\theta^2 m) (1 + \lambda m)}.$$

Taking the difference between the expressions at $m + 1$ and m , placing the two fractions over a common denominator, and simplifying gives

$$L(d+1) - L(d) = \frac{\sigma_\theta^4 [\lambda \sigma_\theta^2 m (m+1) - \sigma_\varepsilon^2]}{(\sigma_\varepsilon^2 + \sigma_\theta^2 m) (1 + \lambda m) (\sigma_\varepsilon^2 + \sigma_\theta^2 (m+1)) (1 + \lambda (m+1))}.$$

Indeed, the numerator generated by the difference of the two fractions is

$$\begin{aligned} m (\sigma_\varepsilon^2 + \sigma_\theta^2 (m+1)) (1 + \lambda (m+1)) - (m+1) (\sigma_\varepsilon^2 + \sigma_\theta^2 m) (1 + \lambda m) \\ = \lambda \sigma_\theta^2 m (m+1) - \sigma_\varepsilon^2. \end{aligned}$$

If $\lambda = 0$, the denominator is positive and the numerator is $-\sigma_\varepsilon^2 < 0$, so $L(d+1) < L(d)$ for every d . This proves part (i).

Suppose $\lambda > 0$. The denominator in the increment formula is strictly positive, so the sign of $L(d+1) - L(d)$ is the sign of

$$m(m+1) - \frac{\sigma_\varepsilon^2}{\lambda\sigma_\theta^2}.$$

The map $x \mapsto x(x+1)$ is strictly increasing on $(0, \infty)$, and its unique positive solution to

$$x(x+1) = \frac{\sigma_\varepsilon^2}{\lambda\sigma_\theta^2}$$

is

$$x_\lambda = \frac{\sqrt{1 + 4\sigma_\varepsilon^2/(\lambda\sigma_\theta^2)} - 1}{2}.$$

Therefore the increments are negative when $m < x_\lambda$, zero when $m = x_\lambda$, and positive when $m > x_\lambda$. This proves that $L(d)$ is single-troughed.

If $x_\lambda \notin \mathbb{N}$, let $n := \lceil x_\lambda \rceil$. Then the increments are negative for every integer $m < n$ and positive for every integer $m \geq n$, so the unique minimizer is $d = n - 1 = \lceil x_\lambda \rceil - 1$. If $x_\lambda = n \in \mathbb{N}$, the increments are negative for $m < n$, zero for $m = n$, and positive for $m > n$. Hence $d = n - 1$ and $d = n$ are the two minimizers. Equivalently, these adjacent dimensions are tied exactly when

$$\lambda = \frac{\sigma_\varepsilon^2}{\sigma_\theta^2 n(n+1)}.$$

In particular, $d = 0$ is the unique minimizer when $x_\lambda < 1$, equivalently when $\lambda > \sigma_\varepsilon^2/(2\sigma_\theta^2)$, and $d = 0$ and $d = 1$ are tied when equality holds.

For completeness, the continuous extension

$$\tilde{L}(m) := \sigma_\theta^2 - \frac{\sigma_\theta^4 m}{(\sigma_\varepsilon^2 + \sigma_\theta^2 m)(1 + \lambda m)}$$

satisfies

$$\tilde{L}'(m) = -\frac{\sigma_\theta^4 (\sigma_\varepsilon^2 - \lambda\sigma_\theta^2 m^2)}{(\sigma_\varepsilon^2 + \sigma_\theta^2 m)^2 (1 + \lambda m)^2}.$$

Thus its unconstrained minimizer over $m > 0$ is $\sqrt{\sigma_\varepsilon^2/(\lambda\sigma_\theta^2)}$, while its minimizer over the feasible continuous domain $m \geq 1$ is

$$\max \left\{ 1, \sqrt{\frac{\sigma_\varepsilon^2}{\lambda\sigma_\theta^2}} \right\}.$$

■

Proof of Proposition 8

Fix $d \in \mathbb{N}_0$ and write $m := d+1$. Under Assumption 7, the state and signal vector are jointly Gaussian, so Assumption 4 holds. The quadratic processing cost satisfies Assumptions 3 and 5. Proposition 4 therefore allows us to restrict attention to estimates $\ell(S^{(d)}) = \omega^\top S^{(d)} + c$.

As in the proof of Proposition 5, the optimal intercept is $c = (1 - \omega^\top \mathbf{1}_m) \mu_\theta$. After centering, the objective becomes

$$\sigma_\theta^2 - 2\sigma_\theta^2 \omega^\top \mathbf{1}_m + (1 + \lambda m) \omega^\top \Sigma \omega,$$

where

$$\Sigma = \sigma_\theta^2 \mathbf{1}_m \mathbf{1}_m^\top + D, \quad D := \text{diag}(\sigma_{\varepsilon,0}^2, \dots, \sigma_{\varepsilon,d}^2).$$

The matrix Σ is positive definite, so the unique minimizer satisfies

$$\omega = \frac{\sigma_\theta^2}{1 + \lambda m} \Sigma^{-1} \mathbf{1}_m.$$

The Sherman–Morrison formula gives

$$\Sigma^{-1} = D^{-1} - \frac{\sigma_\theta^2 D^{-1} \mathbf{1}_m \mathbf{1}_m^\top D^{-1}}{1 + \sigma_\theta^2 \mathbf{1}_m^\top D^{-1} \mathbf{1}_m}.$$

Since $P_d = \mathbf{1}_m^\top D^{-1} \mathbf{1}_m$,

$$\Sigma^{-1} \mathbf{1}_m = D^{-1} \mathbf{1}_m \left(1 - \frac{P_d}{1/\sigma_\theta^2 + P_d} \right) = \frac{1}{1 + \sigma_\theta^2 P_d} D^{-1} \mathbf{1}_m.$$

Consequently,

$$\omega_{d,k}^* = \frac{1}{1 + \lambda m} \frac{\sigma_\theta^2}{1 + \sigma_\theta^2 P_d} \frac{1}{\sigma_{\varepsilon,k}^2}, \quad k = 0, 1, \dots, d,$$

and $c_d^* = (1 - (\omega_d^*)^\top \mathbf{1}_m) \mu_\theta$. The factor multiplying $1/\sigma_{\varepsilon,k}^2$ is positive and common to all coordinates. Hence

$$\sigma_{\varepsilon,k}^2 < \sigma_{\varepsilon,j}^2 \iff \omega_{d,k}^* > \omega_{d,j}^*, \quad \frac{\omega_{d,k}^*}{\omega_{d,j}^*} = \frac{1/\sigma_{\varepsilon,k}^2}{1/\sigma_{\varepsilon,j}^2} = \frac{\sigma_{\varepsilon,j}^2}{\sigma_{\varepsilon,k}^2}.$$

■

Proof of Proposition 9

Write $m := d + 1$ and abbreviate the induced estimates $\ell_d^A(S^{(d)})$ and $\ell_d^N(S^{(d)})$ by ℓ_d^A and ℓ_d^N . Under the true correlated-Gaussian law,

$$\Sigma_m^A := \text{Var}(S^{(d)}) = \sigma_\varepsilon^2(1 - \rho)I_m + (\sigma_\theta^2 + \rho\sigma_\varepsilon^2)\mathbf{1}_m\mathbf{1}_m^\top.$$

The same first-order condition used in the proof of Proposition 5 gives

$$\omega_m^A = \frac{\sigma_\theta^2}{1 + \lambda m}(\Sigma_m^A)^{-1}\mathbf{1}_m.$$

Since

$$\Sigma_m^A \mathbf{1}_m = m(\sigma_\theta^2 + \sigma_\varepsilon^2 q_m) \mathbf{1}_m,$$

we obtain

$$\omega_m^A = \frac{\kappa_m^A}{m} \mathbf{1}_m,$$

where κ_m^A is defined in (18). Under the subjective independent-noise law, the same calculation sets $\rho = 0$ and gives

$$\omega_m^N = \frac{\kappa_m^N}{m} \mathbf{1}_m,$$

where κ_m^N is defined in (19).

Now let

$$Z_m := \bar{S}^{(d)} - \mu_\theta, \quad V_m := \text{Var}(Z_m) = \sigma_\theta^2 + \sigma_\varepsilon^2 q_m.$$

Under the true law, $\text{Cov}(Z_m, \theta) = \sigma_\theta^2$. For $\ell_\kappa := \mu_\theta + \kappa Z_m$, the true total loss is

$$\mathcal{J}_m(\ell_\kappa) = \sigma_\theta^2 - 2\sigma_\theta^2 \kappa + (1 + \lambda m)V_m \kappa^2,$$

whose minimizer is $\kappa_m^A = \sigma_\theta^2 / ((1 + \lambda m)V_m)$. Completing the square yields

$$\mathcal{J}_m(\ell_\kappa) = \mathcal{J}_m(\ell_d^A) + (1 + \lambda m)V_m(\kappa - \kappa_m^A)^2.$$

Setting $\kappa = \kappa_m^N$ and recalling $\Delta_m = \kappa_m^N - \kappa_m^A$ gives the exact loss identity

$$\mathcal{J}_m(\ell_d^N) - \mathcal{J}_m(\ell_d^A) = (1 + \lambda m)V_m \Delta_m^2 \geq 0.$$

Moreover,

$$\ell_d^N - \ell_d^A = \Delta_m Z_m, \quad \omega_m^N - \omega_m^A = \frac{\Delta_m}{m} \mathbf{1}_m,$$

so

$$\|\ell_d^N - \ell_d^A\|_2 = |\Delta_m| \sqrt{V_m}, \quad \|\omega_m^N - \omega_m^A\|_E = \frac{|\Delta_m|}{\sqrt{m}},$$

where $\|\cdot\|_E$ denotes the Euclidean norm on coefficient vectors.

Suppose first that $\lambda > 0$. Since $q_m \rightarrow \rho$, the sequence V_m converges to $\sigma_\theta^2 + \rho\sigma_\varepsilon^2$ and is bounded above and away from zero. The formulas for the two coefficients imply

$$\kappa_m^A = O(m^{-1}), \quad \kappa_m^N = O(m^{-1}), \quad \Delta_m = O(m^{-1}).$$

Therefore

$$\|\ell_d^A - \mu_\theta\|_2 = |\kappa_m^A| \sqrt{V_m} \rightarrow 0, \quad \|\ell_d^N - \mu_\theta\|_2 = |\kappa_m^N| \sqrt{V_m} \rightarrow 0.$$

The preceding exact identities also give

$$\|\ell_d^N - \ell_d^A\|_2 = O(m^{-1}), \quad \|\omega_m^N - \omega_m^A\|_E = O(m^{-3/2}),$$

and

$$0 \leq \mathcal{J}_m(\ell_d^N) - \mathcal{J}_m(\ell_d^A) = (1 + \lambda m) V_m \Delta_m^2 = O(m^{-1}).$$

This proves part (i).

Now suppose $\lambda = 0$. Since $q_m \rightarrow \rho$,

$$\kappa_m^A \rightarrow \frac{\sigma_\theta^2}{\sigma_\theta^2 + \rho\sigma_\varepsilon^2}, \quad \kappa_m^N \rightarrow 1, \quad \Delta_m \rightarrow \frac{\rho\sigma_\varepsilon^2}{\sigma_\theta^2 + \rho\sigma_\varepsilon^2}.$$

Because $V_m \rightarrow \sigma_\theta^2 + \rho\sigma_\varepsilon^2$,

$$\mathbb{E}[(\ell_d^N - \ell_d^A)^2] = V_m \Delta_m^2 \rightarrow \frac{\rho^2 \sigma_\varepsilon^4}{\sigma_\theta^2 + \rho\sigma_\varepsilon^2}.$$

When $\lambda = 0$, the exact loss identity reduces to

$$\mathcal{J}_m(\ell_d^N) - \mathcal{J}_m(\ell_d^A) = V_m \Delta_m^2,$$

so the true total-loss gap has the same strictly positive limit. This proves part (ii). ■

References

- Blackwell, D. (1953). “Equivalent comparisons of experiments”. *The Annals of Mathematical Statistics* 24.2, pp. 265–272.
- Blankespoor, E., E. deHaan, and I. Marinovic (2020). “Disclosure Processing Costs, Investors’ Information Choice, and Equity Market Outcomes: A Review”. *Journal of Accounting and Economics* 70.2–3, p. 101344.
- Branco, F., M. Sun, and J. M. Villas-Boas (2016). “Too Much Information? Information Provision and Search Costs”. *Marketing Science* 35.4, pp. 605–618.
- Caplin, A. and M. Dean (2015). “Revealed preference, rational inattention, and costly information acquisition”. *American Economic Review* 105.7, pp. 2183–2203.
- Chadd, I., E. Filiz-Ozbay, and E. Y. Ozbay (2021). “The Relevance of Irrelevant Information”. *Experimental Economics* 24.3, pp. 985–1018.
- Chewning Jr, E. G. and A. M. Harrell (1990). “The effect of information load on decision makers’ cue utilization levels and decision quality in a financial distress decision task”. *Accounting, organizations and society* 15.6, pp. 527–542.
- Dasaratha, K. and K. He (2020). “Network structure and naive sequential learning”. *Theoretical Economics* 15.2, pp. 415–444.
- Dasaratha, K. and K. He (2021). “An experiment on network density and sequential learning”. *Games and Economic Behavior* 128, pp. 182–192.
- DellaVigna, S. and J. M. Pollet (2009). “Investor Inattention and Friday Earnings Announcements”. *The Journal of Finance* 64.2, pp. 709–749.
- Dummel, S., J. Rummel, and A. Voss (2016). “Additional information is not ignored: New evidence for information integration and inhibition in take-the-best decisions”. *Acta psychologica* 163, pp. 167–184.
- Enke, B. and F. Zimmermann (2019). “Correlation neglect in belief formation”. *The review of economic studies* 86.1, pp. 313–332.
- Eppler, M. J. and J. Mengis (2004). “The concept of information overload: A review of literature from organization Science, accounting, marketing, MIS, and related disciplines”. *The Information Society* 20.5, pp. 325–344.
- Esponda, I. and D. Pouzo (2016). “Berk–Nash equilibrium: A framework for modeling agents with misspecified models”. *Econometrica* 84.3, pp. 1093–1130.
- Frick, M., R. Iijima, and Y. Ishii (2020). “Misinterpreting Others and the Fragility of Social Learning”. *Econometrica* 88.6, pp. 2281–2328.
- Gabaix, X. (2014). “A sparsity-based model of bounded rationality”. *The Quarterly Journal of Economics* 129.4, pp. 1661–1710.

- Grossman, S. J. and J. E. Stiglitz (1980). “On the impossibility of informationally efficient markets”. *The American Economic Review* 70.3, pp. 393–408.
- Hirshleifer, D., S. S. Lim, and S. H. Teoh (2009). “Driven to Distraction: Extraneous Events and Underreaction to Earnings News”. *The Journal of Finance* 64.5, pp. 2289–2325.
- Hirshleifer, D. and S. H. Teoh (2003). “Limited Attention, Information Disclosure, and Financial Reporting”. *Journal of Accounting and Economics* 36.1–3, pp. 337–386.
- Iyengar, S. S. and M. R. Lepper (2000). “When choice is demotivating: Can one desire too much of a good thing?” *Journal of Personality and Social Psychology* 79.6, pp. 995–1006.
- Jacoby, J., D. E. Speller, and C. A. Kohn (1974). “Brand choice behavior as a function of information load”. *Journal of marketing research* 11.1, pp. 63–69.
- Lleras, J. S., Y. Masatlioglu, D. Nakajima, and E. Y. Ozbay (2017). “When More Is Less: Limited Consideration”. *Journal of Economic Theory* 170, pp. 70–85.
- Lu, J. (2022). “Limited Attention: Implications for Financial Reporting”. *Journal of Accounting Research* 60.5, pp. 1991–2027.
- Malhotra, N. K. (1982). “Information load and consumer decision making”. *Journal of Consumer Research* 8.4, pp. 419–430.
- Masatlioglu, Y., D. Nakajima, and E. Y. Ozbay (2012). “Revealed Attention”. *American Economic Review* 102.5, pp. 2183–2205.
- Matějka, F. and A. McKay (2015). “Rational inattention to discrete choices: A new foundation for the multinomial logit model”. *American Economic Review* 105.1, pp. 272–298.
- Payne, J. W. (1976). “Task complexity and contingent processing in decision making: An information search and protocol analysis”. *Organizational behavior and human performance* 16.2, pp. 366–387.
- Rodriguez, M. G., K. Gummadi, and B. Schoelkopf (2014). “Quantifying information overload in social media and its impact on social contagions”. In: *Proceedings of the international AAAI conference on web and social media*. Vol. 8. 1, pp. 170–179.
- Roetzel, P. G. (2019). “Information overload in the information age: a review of the literature from business administration, business psychology, and related disciplines with a bibliometric approach and framework development”. *Business research* 12.2, pp. 479–522.
- Sims, C. A. (2003). “Implications of rational inattention”. *Journal of monetary Economics* 50.3, pp. 665–690.
- Spiegler, R. (2016). “Bayesian Networks and Boundedly Rational Expectations”. *The Quarterly Journal of Economics* 131.3, pp. 1243–1290.
- Vives, X. (1993). “How fast do rational agents learn?” *The Review of Economic Studies* 60.2, pp. 329–347.